0 FASTQ entry

clinical FASTQs, paired or tumour-only

FASTQC → fgbio UMI → BWA-MEM2 → MarkDuplicates → BQSR

1 Clinical inputs and startup checks

Sample sheet
patient, sample, status
BAM or FASTQ; tumour-only ok

Processed vendor BAMs
b37 or chr contig names
enter at variant_calling

Capture targets
--targets BED, or
inferred from coverage

Startup checks
sheet, type, contigs
→ paired, tumour-only

2 BAM harmonisation, targets and per-assay resources

17 → chr17

RENAME_BAM_HEADER ADAPTED
contig rename b37 ↔ chr when lengths match
name, length and order checked against --ref
inputs staged apart; originals untouched

PREPARE_TARGETS NEW
pad ±100 bp, merge → Strelka2 call regions
20 interval shards → Mutect2, pileups
clean BED → mosdepth, CNVkit, BQSR

Built once per assay from the cohort's normals: CNVkit reference · PureCN normal DB · Mutect2 cohort PON

3 Paired tumour–normal calling

Somatic callers

Mutect2 · Strelka2 · MuSE · SAGE

adapted: Strelka2 call regions, panel contamination fallback

QC and copy number

Conpair · mosdepth · CNVkit (pooled) · PureCN

ASCAT · Delly · Manta — genome-wide; off for panels

4 Tumour-only calling NEW

Somatic callers

Mutect2 tumour-only · SAGE tumour-only

Mutect2 with panel of normals + gnomAD; SAGE without -reference

QC and copy number

mosdepth · CNVkit (flat or pooled) · PureCN

Need a normal: Strelka2 · MuSE · Conpair · ASCAT · Delly · Manta

5 Consensus calls, biomarkers and reports

Paired: POST_SCENARIOS (upstream)
scenarios 1–6 · final VCF + commands
no-consensus report · AlleleCounter VAF

Tumour-only consensus
PASS in both Mutect2 and SAGE
germline_popaf if population AF > 0.1%

--outdir
VCFs, CNVkit, PureCN
purity, ploidy, LOH

VEP + vcf2maf · TMB / Mb · MSI · gene status · somalier · MultiQC · PCGR report

New in Clinical SomaticBunny · Original step used for clinical data (some adapted) · Original step not used for clinical data